

File permissions in Linux

Project description

In this activity, I worked as a security professional responsible for managing file permissions for a research team. I examined file and directory permissions using Linux commands and modified them to remove unauthorized access. This helped improve system security and enforce proper authorization, and protect sensitive files. I used the `ls -la` command to view permissions and the `chmod` command to change permissions for files and directories

Check file and directory details

The following code demonstrates how I used Linux commands to determine the existing permissions set for a specific directory in the file system.

Command used

```
ls -la
```

Explanation

I used the `ls -la` command to display all files and directories, including hidden files, and to view their permissions.

```
researcher2@09235e9837aa:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Jun 23 12:22 .
drwxr-xr-x 3 researcher2 research_team 4096 Jun 23 12:47 ..
-r--r----- 1 researcher2 research_team  46 Jun 23 12:22 .project_x.txt
drwx----- 2 researcher2 research_team 4096 Jun 23 12:22 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Jun 23 12:22 project_k.txt
-rw----- 1 researcher2 research_team  46 Jun 23 12:22 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 23 12:22 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 23 12:22 project_t.txt
```

Current Permissions

The permission string is a 10 -character string that shows the permissions assigned to a file or directory

Describe the permissions string

The permission string `-rw-rw-r--` contains 10 characters

The first character (`-`) shows that it is a regular file. .

The next three characters (`rw-`) show the owner's permissions. The owner can read and write the file.

The next three characters (`rw-`) show the group's permissions. The group can read and write the file.

The last three characters (`r--`) show the permissions for others. Others can only read the file.

Explanation

This string shows who can access the file and what actions they can perform. I removed write permission for others.

Change file permissions

Command used

`chmod o-w project_k.txt`

```
researcher2@09235e9837aa:~/projects$ chmod o-w project_k.txt
researcher2@09235e9837aa:~/projects$ ls -l
total 20
drwx----- 2 researcher2 research_team 4096 Jun 23 12:22 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Jun 23 12:22 project_k.txt
-rw----- 1 researcher2 research_team  46 Jun 23 12:22 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 23 12:22 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 23 12:22 project_t.txt
```

Explanation

The organization does not allow others to have write access to files. I used the `chmod` command to remove write permission from others for the file `project_k.txt`

After running the command, the file permissions remained:

`-rw-rw-r--` which confirms that others do not have write access.

Result

The file no longer allows others to modify its content

Change file permissions on a hidden file

Command used: `chmod u-w,g-w,g+r .project_x.txt`

```
researcher2@09235e9837aa:~/projects$ chmod u-w,g-w,g+r .project_x.txt
researcher2@09235e9837aa:~/projects$ ls -l
total 20
drwx----- 2 researcher2 research_team 4096 Jun 23 12:22 drafts
-rw-rw-r-- 1 researcher2 research_team 46 Jun 23 12:22 project_k.txt
-rw----- 1 researcher2 research_team 46 Jun 23 12:22 project_m.txt
-rw-rw-r-- 1 researcher2 research_team 46 Jun 23 12:22 project_r.txt
-rw-rw-r-- 1 researcher2 research_team 46 Jun 23 12:22 project_t.txt
```

Explanation:

Ensured the hidden file could only be read by the user and group. The hidden file `.project_x.txt` should not allow write access. The user and group should only be able to read the file. I used this command to remove write permissions and ensure the group had read access.

Output

The permissions became:

`-r--r-----`

Result

The file can be read by the user and group, but it cannot be modified

Change directory permissions

Command used

`chmod g-x drafts`

```
researcher2@9180ca370360:~/projects$ chmod g-x drafts
researcher2@9180ca370360:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Jul 6 21:46 .
drwxr-xr-x 3 researcher2 research_team 4096 Jul 6 23:20 ..
-r--r----- 1 researcher2 research_team 46 Jul 6 21:46 .project_x.txt
drwx----- 2 researcher2 research_team 4096 Jul 6 21:46 drafts
-rw-rw-r-- 1 researcher2 research_team 46 Jul 6 21:46 project_k.txt
-rw----- 1 researcher2 research_team 46 Jul 6 21:46 project_m.txt
-rw-rw-r-- 1 researcher2 research_team 46 Jul 6 21:46 project_r.txt
-rw-rw-r-- 1 researcher2 research_team 46 Jul 6 21:46 project_t.txt
researcher2@9180ca370360:~/projects$
```

Explanation:

Removed execute permission from the group on the drafts directory. Only the owner (researcher2) should access the drafts directory. I used the `chmod` command to remove execute permission from the group for the `drafts` directory.

Result

Group members can no longer access the contents of the drafts directory.

Summary

In this research activity, I used Linux commands such as `ls -la` to view file permissions and `chmod` to update permissions on file and directories. I identified insecure permissions and corrected them by removing unnecessary write and execute access. These changes improved the security of the research team's files and directories, I also learned how to read the 10-character permission string and manage permissions for hidden files. These changes helped ensure that only authorized users could access or modify important files.